

Zachary Lin

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EDUCATION

Rensselaer Polytechnic Institute

Computer Science Major

- GPA: 3.88
- Relevant Coursework: Data Structures, Discrete Mathematics, Operating Systems, Computer Architecture

Troy, NY

August 2025 – Present

EXPERIENCE

Software Engineer Intern

Bridges To Science

March 2025 – July 2025

Remote

- Streamlined organizational communications and reduced manual content creation time by **60%** by building a **RAG-based chatbot** that generates personalized emails, flyers, and templates using document processing and dynamic data scraping.
- Eliminated repetitive manual email drafting and improved communication clarity by automating **100+ dynamic, personalized emails** using Google Apps Script and Google Sheets.
- Improved system efficiency and maintained response quality by **doubling request throughput and reducing operational costs** through Gemini-2.0 Flash implementation.

Community Intern

Chinatown East Neighborhood Council

June 2024 – August 2024

New York City, New York

- Drove internship outreach and engagement by leading development of a website highlighting past activities and organizational details, generating **50+ visitors at launch** and enabling prospective intern sign-ups.

Software Engineer Intern

Nagarro

April 2024 – May 2024

New York City, New York

- Improved study efficiency and information accessibility by developing an AI-powered tool with colleagues that leverages **RAG** to **summarize up to 10 videos and documents simultaneously**.
- Collaborated with designers to optimize UI/UX, delivering a cleaner, more accessible interface and improving user navigation.
- Led a live demo and presentation of our application to **20+ NYCDOE officials**, translating complex technical features for a non-technical audience.

PROJECTS

Ticketing System/File Drive | Javascript, Typescript, Expressjs, Reactjs, MongoDB

June 2025 – August 2025

- Developed a ticketing system and file drive with a self-hosted database served over TCP/IP and secured via Cloudflare.
- Reduced query latency by **40%** by implementing database indexing and increasing virtual machine memory allocation, minimizing disk I/O and significantly improving application response times.
- Improved application performance by **reducing server requests 3×** through client-side **debouncing**.

Animal Waste Notification App | Swift, SwiftUI

May 2025

- Earned **2nd place out of 12 teams** by presenting a live demo to Apple SWE judges, showcasing our app's technical innovation and design.
- Developed an iOS app leveraging **14k+ public records** and user reports to notify users of nearby animal waste, encouraging community participation.

Next Mobile Ventures | Python, Javascript, SASS, Reactjs, Nextjs, FastAPI, MongoDB

June 2024 – August 2024

- Streamlined access to car market valuations for **100k+ models** by developing a web app that aggregates data from 3 external APIs, improving user decision-making.
- Improved user experience by reducing average page load time by **50%** through Next.js **Static Site Generation** and browser caching of previously searched results.

TECHNICAL SKILLS

Languages: HTML/CSS, JavaScript, Typescript, SQL, Go, C, C++, Python

Frameworks: Expressjs, FastAPI, Nextjs, Vite, Bootstrap

Data: MongoDB, PostgreSQL, MySQL, Firebase, SQLC, Redis

Cloud/Tooling: Makefile, Git, GitHub, Postman, Docker, Linux, Azure